Tutorial 11

Trees and networks

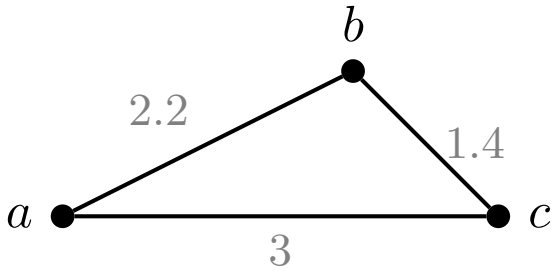
Homework

1. Maximum Flow and Minimum Cut: Select any three networks from Q3 (Parts 1–10) and provide detailed solutions.
2. Breadth-First and Depth-First Search: Complete Q5 (Parts (i)–(iv)) for both graphs (Q5–1 and Q5–2). Provide detailed solutions.

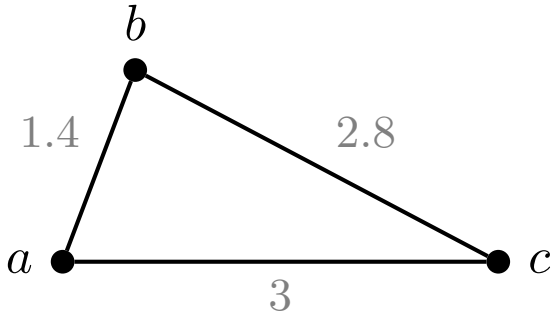
Question 1. For each of the following triangles

- Perform Torricelli's Construction to find the Fermat point. It is recommended that you use a protractor (or a compass) and a ruler.
- Compute the total length of the shortest network connecting the vertices of the triangle.

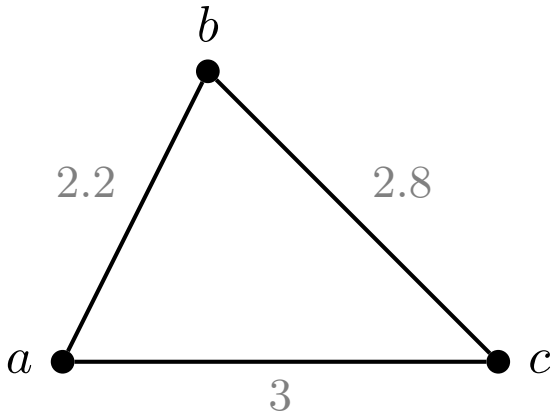
1.



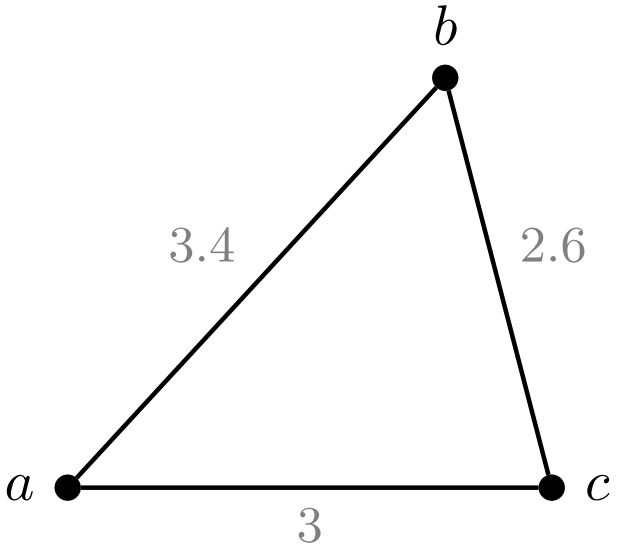
2.



3.



4.



Solution.

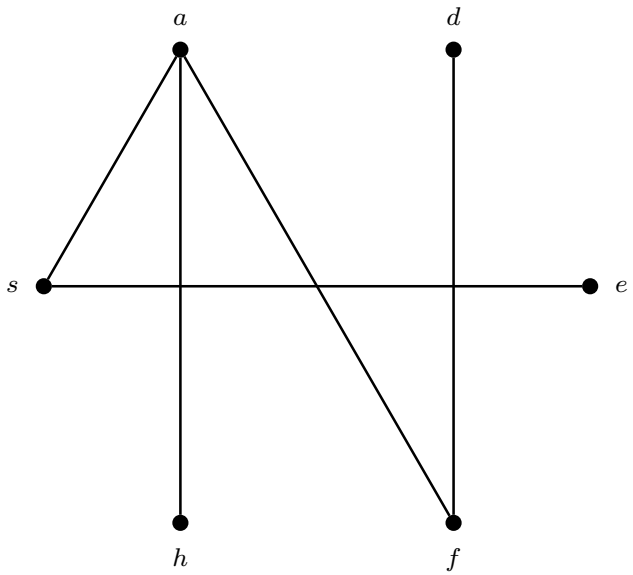
1. 3.633
2. 4.044
3. 4.625
4. 5.206

Question 2. Alice must visit clients in six cities next month and needs to minimize her driving mileage. The table below lists the distances between these cities. Use the mTSP Algorithm to find a good travel plan if she must start and end her trip in Dallas. Compute the total distance.

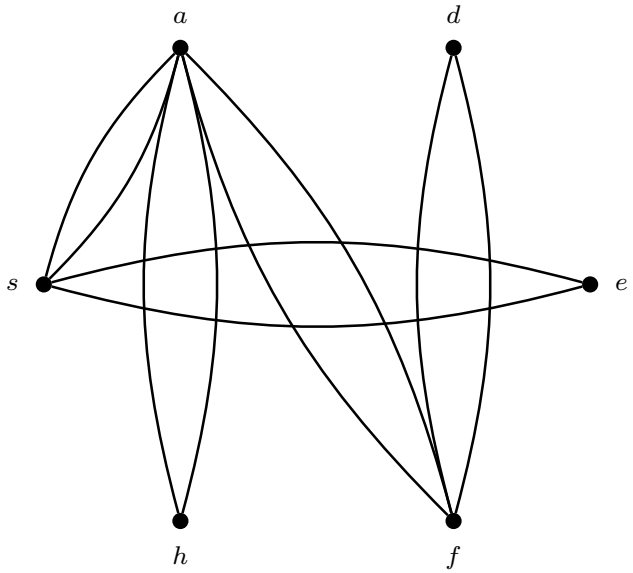
	Austin	Dallas	El Paso	Fort Worth	Houston	San Antonio
Austin	.	182	526	174	146	74
Dallas	182	.	568	31	225	253
El Paso	526	568	.	537	672	500
Fort Worth	174	31	537	.	237	241
Houston	146	225	672	237	.	189
San Antonio	74	253	500	241	189	.

Solution.

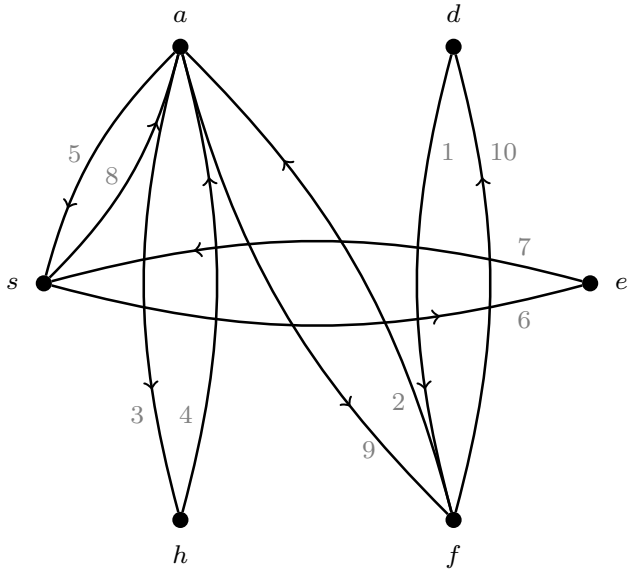
- Step 1. Find a minimum spanning tree (for example, by Prim's Algorithm).



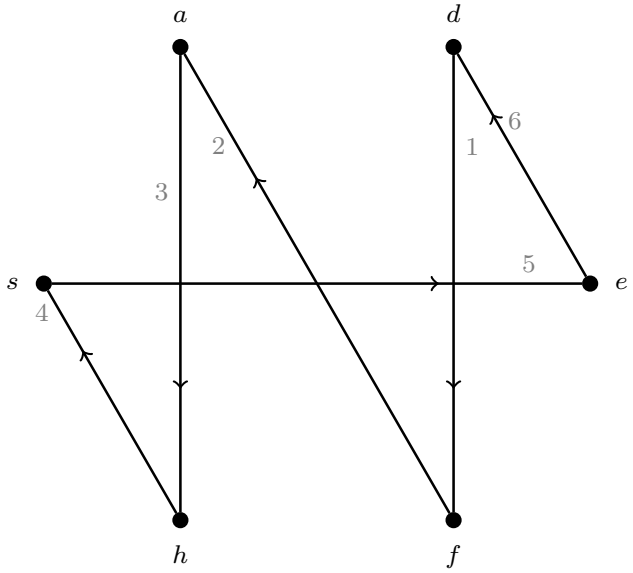
- Step 2. Duplicate all edges of the MST.



- Step 3. Find an Eulerian circuit.



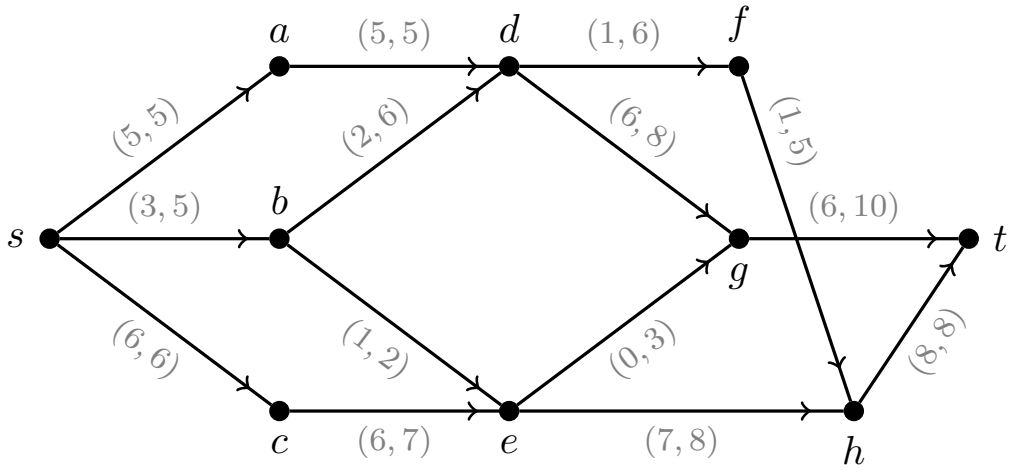
- Step 4. Convert the Eulerian circuit into a Hamiltonian cycle.



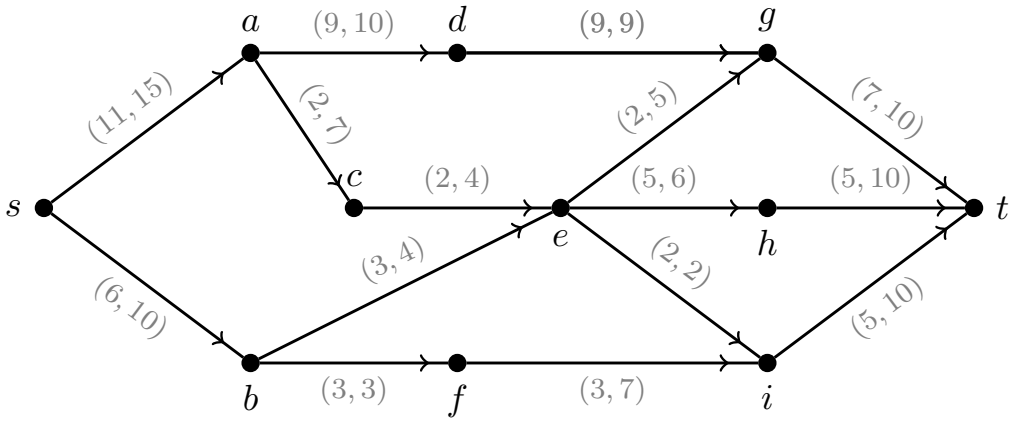
- Step 5. The total distance is 1608.

Question 3. For each of the networks below, use the Augmenting Flow Algorithm to maximize the flow and the Min-Cut Method to find a minimum cut.

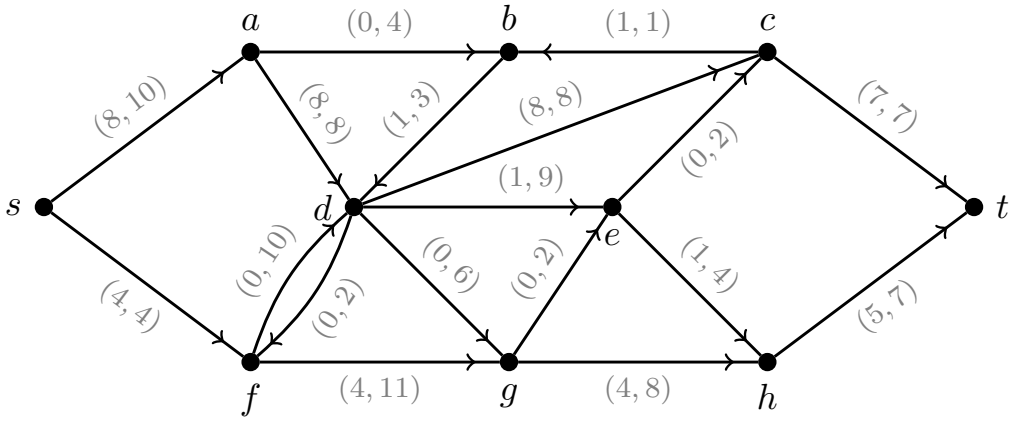
1.



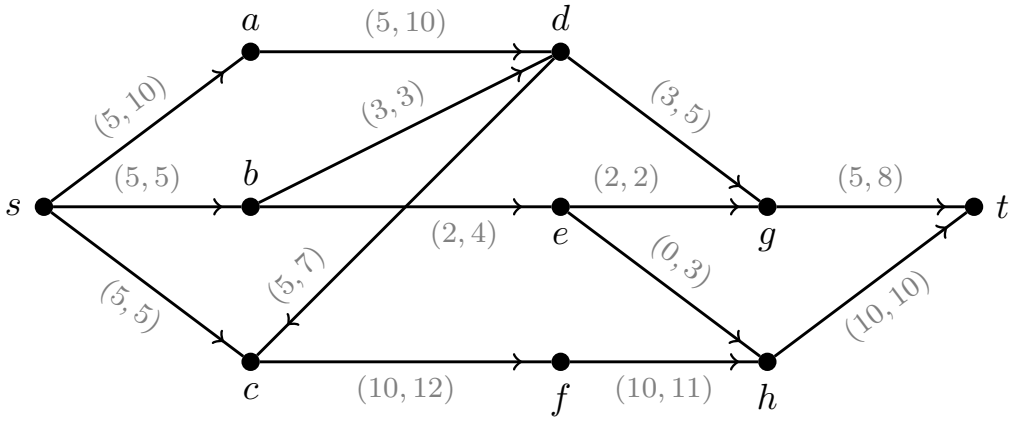
2.



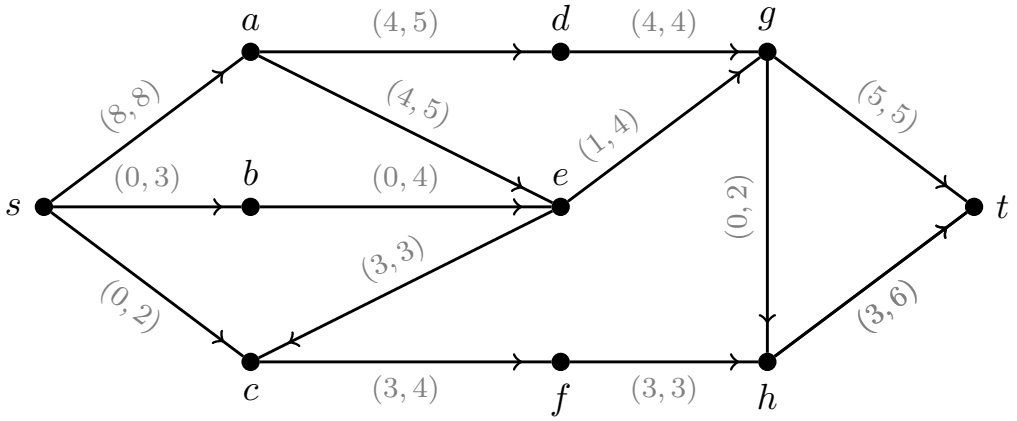
3.



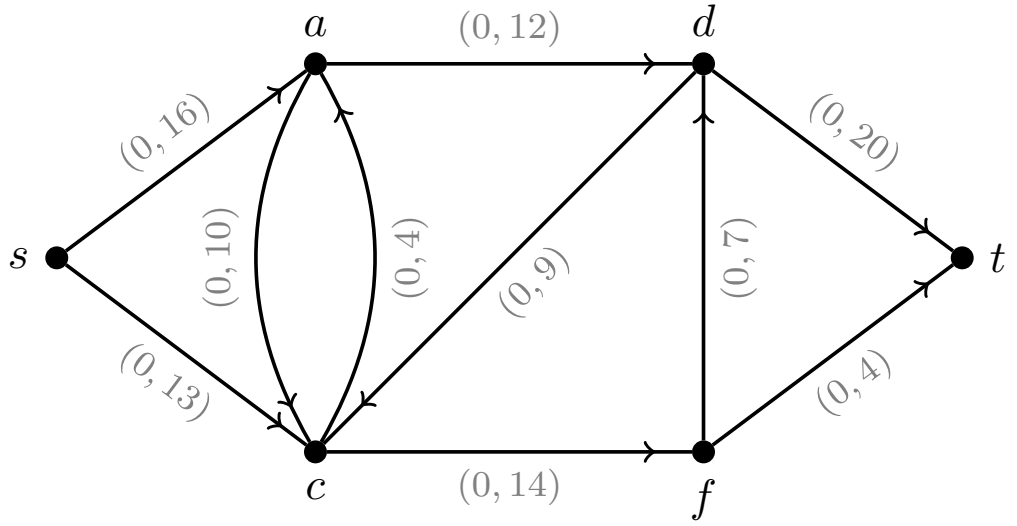
4.


a
(5, 10)
d
(5, 10)
(3, 3)
(3, 5)
(5, 5)
b
(2, 2)
(5, 8)
s
t
(2, 4)
e
g
(0, 3)
(5, 5)
(5, 7)
(10, 10)
c
(10, 12)
f
(10, 11)
h

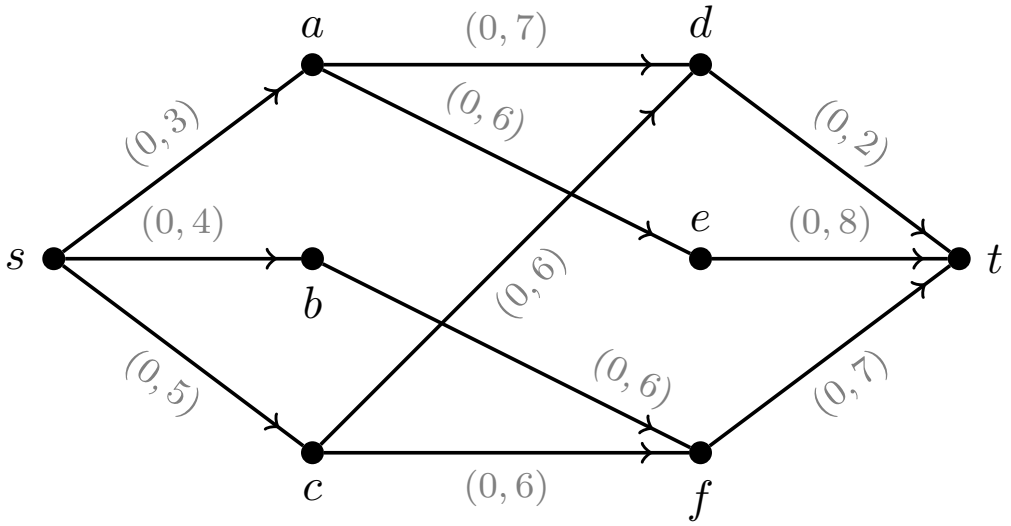
5.


a
(4, 5)
d
(4, 4)
g
(8, 8)
(4, 5)
(1, 4)
(5, 5)
(0, 3)
b
(0, 4)
e
s
t
(0, 2)
(3, 3)
(0, 2)
(3, 6)
c
(3, 4)
f
(3, 3)
h

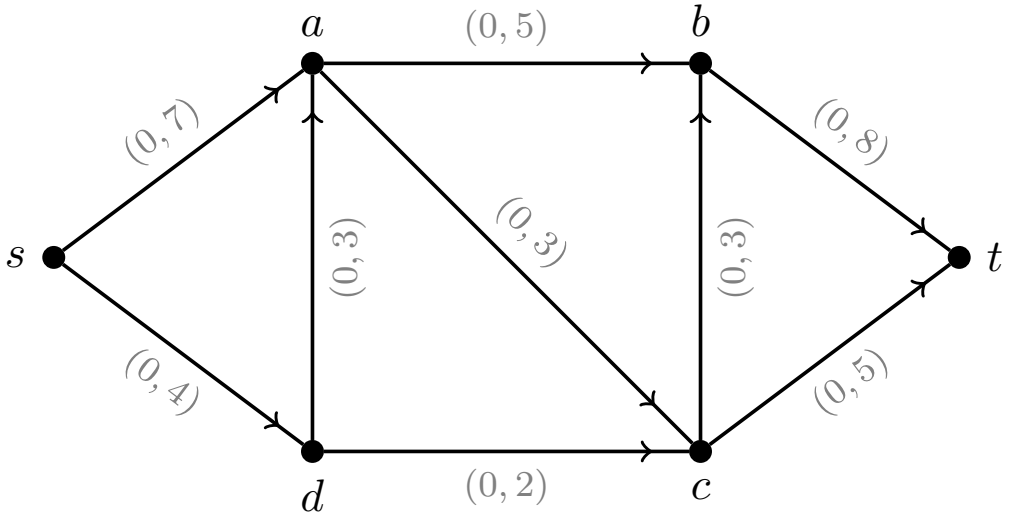
6.


a
(0, 12)
d
(0, 16)
(0, 20)
(0, 10)
(0, 4)
(0, 9)
(0, 7)
s
t
(0, 13)
(0, 4)
c
(0, 14)
f

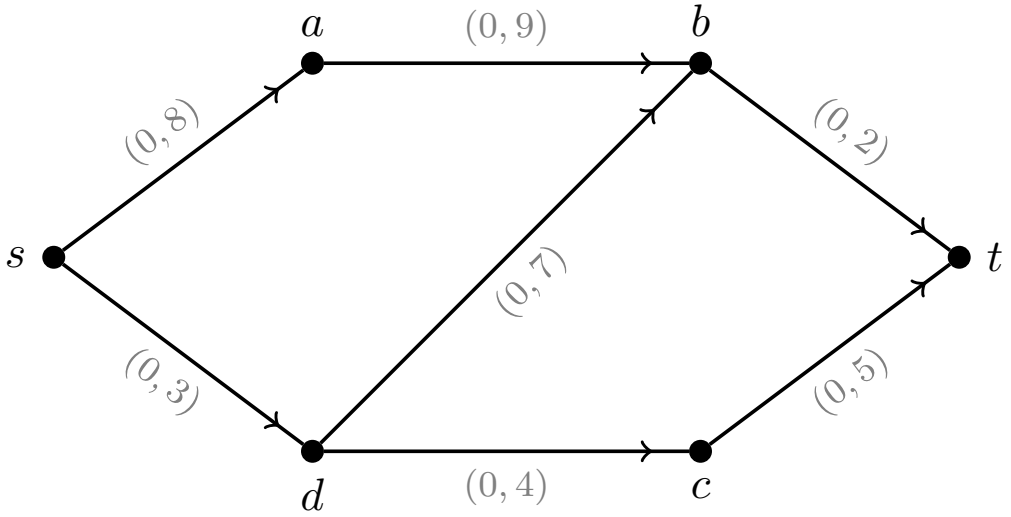
7.


a
(0, 7)
d
(0, 3)
(0, 6)
(0, 2)
(0, 4)
e
(0, 8)
s
t
b
(0, 6)
(0, 5)
(0, 6)
(0, 7)
c
(0, 6)
f

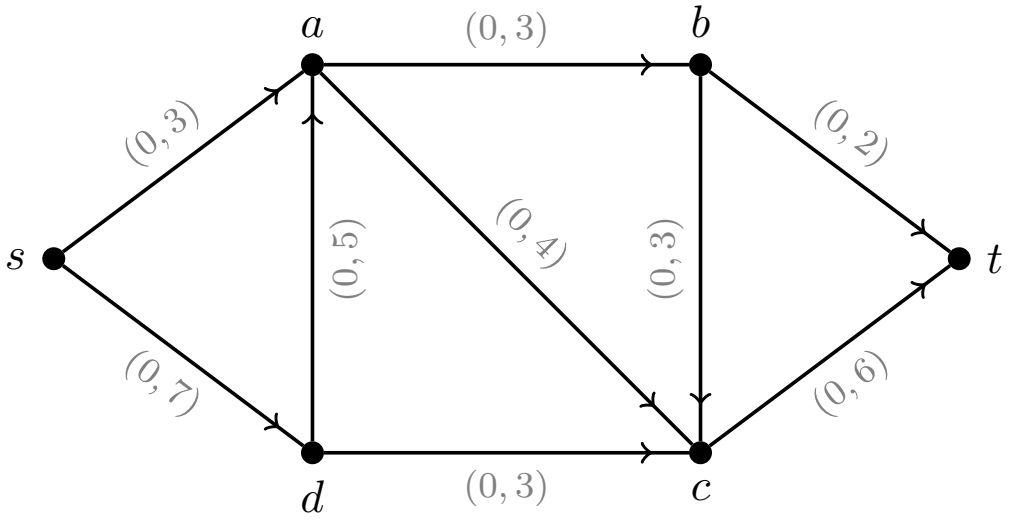
8.


a
(0, 5)
b
(0, 7)
(0, 8)
s
t
(0, 3)
(0, 3)
(0, 3)
(0, 4)
(0, 5)
d
(0, 2)
c

9.


a
(0, 9)
b
(0, 8)
(0, 2)
s
t
(0, 7)
(0, 3)
(0, 5)
d
(0, 4)
c

10.

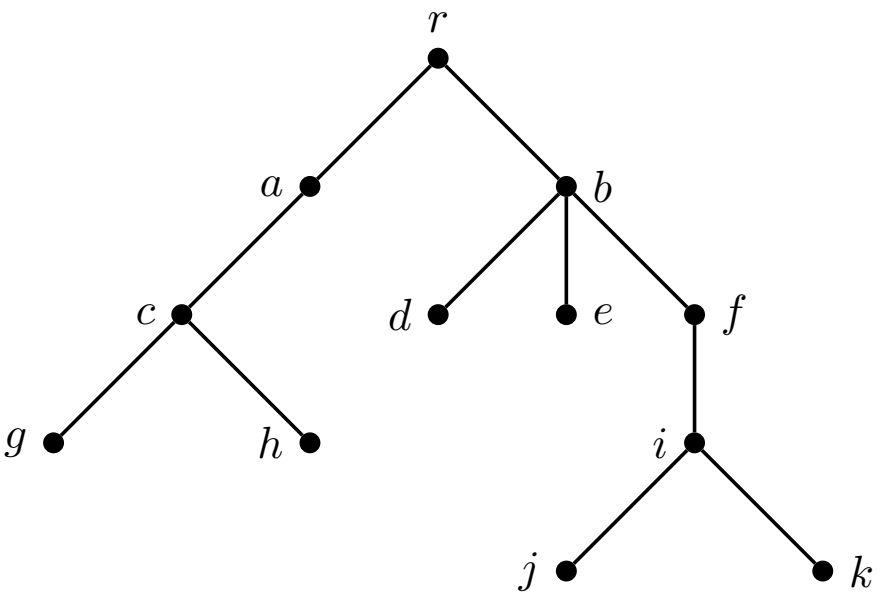


Solution.

1. Maximum flow = 16, $P = \{s\}$
2. Maximum flow = 20, $P = \{s, a, b, c, d\}$
3. Maximum flow = 14, $P = \{s\}$
4. Maximum flow = 17, $P = \{s, a, b, c, d, e, f, h\}$
5. Maximum flow = 10, $P = \{s, a, b, c, d, e, f, g\}$
6. Maximum flow = 23, $P = \{s, a, c, f\}$
7. Maximum flow = 12, $P = \{s, d, c, b, f\}$
8. Maximum flow = 10, $P = \{s, a, d\}$
9. Maximum flow = 5, $P = \{s, a, b\}$
10. Maximum flow = 8, $P = \{s, a, b, c, d\}$

Question 4. For the rooted tree T below, with root r, identify the following:

1. The levels of r, f, h, and k.
2. The height of T.
3. The parents of r, b, c, f, and i.
4. The children of r, b, c, f, and i.
5. The ancestors of r, a, d, h, and k.
6. The descendants of r, a, d, h, and k.
7. The siblings of a, f, h, and i.



Solution.

1. The levels are $0, 2, 3$, and 4, respectively.
2. 4
3. The parents are: none; r; a; b; f, respectively.
4. The children are: a, b; d, e, f; g, h; i; j, k, respectively.
5. The ancestors are: none; r; r, b; r, a, c; r, b, f, i, respectively.
6. The descendants are: $a, b, c, d, e, f, g, h, i, j, k$; c, g, h; none; none; none, respectively.
7. The siblings are: b; d, e; g; none, respectively.

Question 5. Complete each of the following on the two graphs shown below.

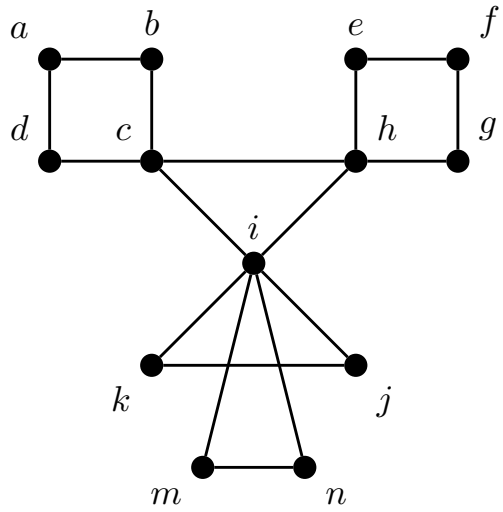
i) Find the breadth-first search tree with root a.

ii) Find the breadth-first search tree with root i.

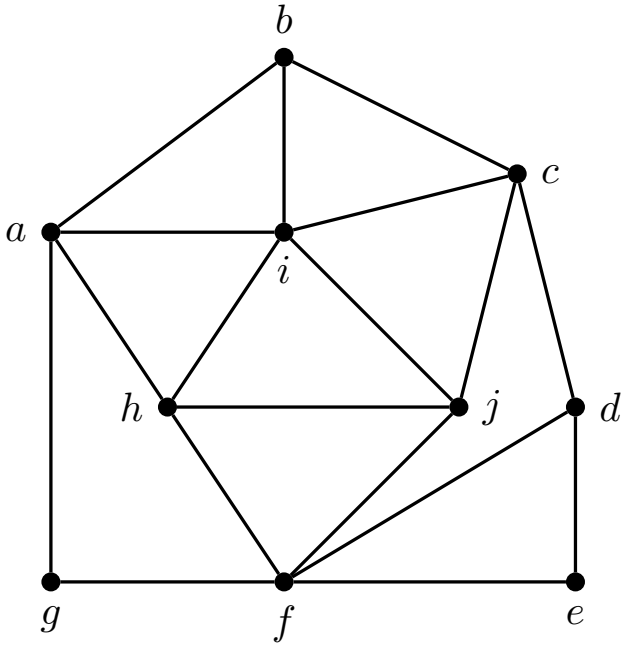
iii) Find the depth-first search tree with root a.

iv) Find the depth-first search tree with root i.

1.

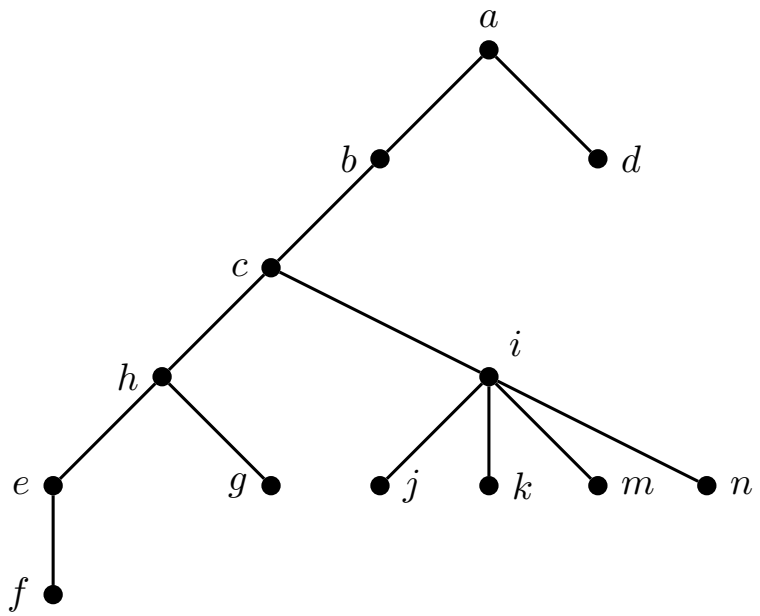


2.

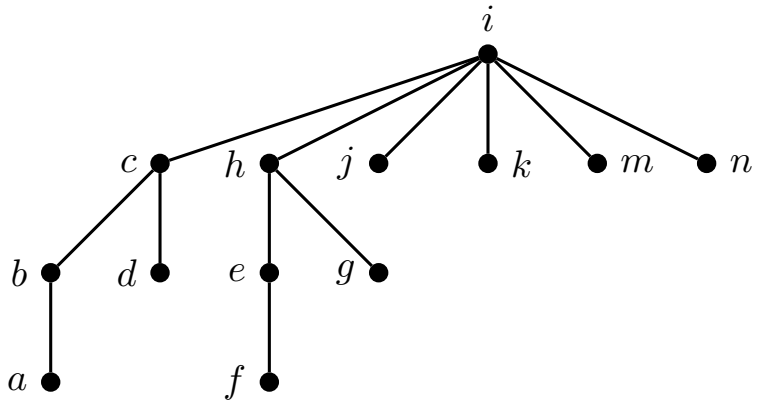


Solution.

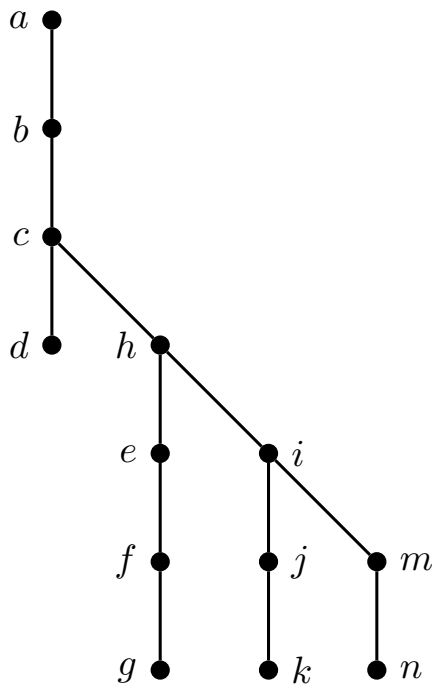
1. i) BFS tree with root a



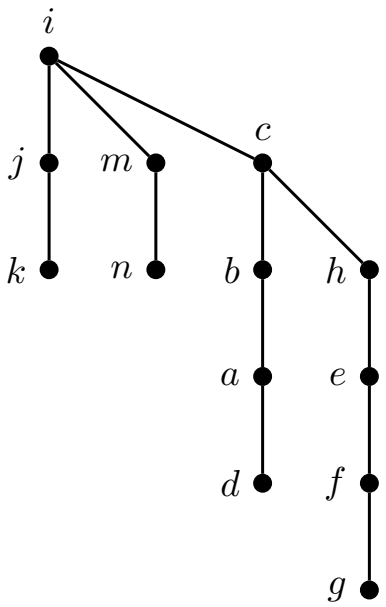
ii) BFS tree with root i



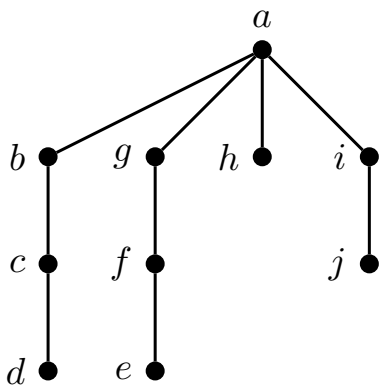
iii) DFS tree with root a



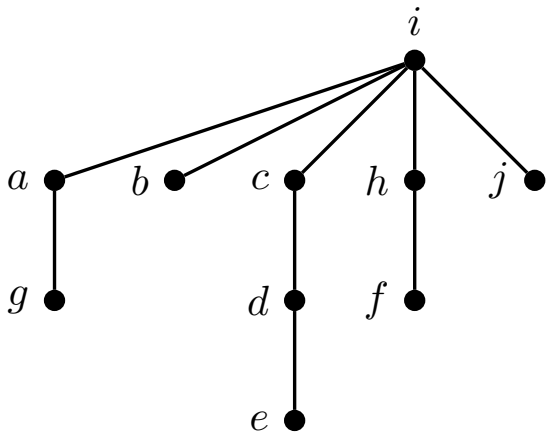
iv) DFS tree with root i



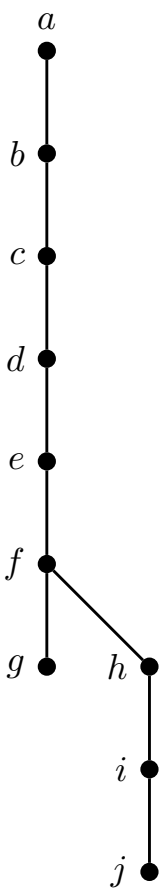
2. i) BFS tree with root a



ii) BFS tree with root i



iii) DFS tree with root a



iv) DFS tree with root i

